

TRUST-AWARE BIO-INSPIRED SWARM DEFENSE

PUNARVITHA

ROLL No: S2020200361

KOMMANA SRINIVASA RAO

ROLL No: S20230030386

VARTHALA KHYATHI YADAV

ROLL No: S20230030418

GUIDED BY: DR SAKTHI SWARRUP J

MAY 2026

PLAN OF THE PRESENTATION

- 1 Project Overview
- 2 Methodology and System Design
- 3 Implementation Progress
- 4 Simulation Setup
- 5 Results
- 6 Novelty and Research Contribution
- 7 Publication Readiness
- 8 Challenges and Future Work

PROJECT OVERVIEW

PROBLEM STATEMENT

- Drone swarms operating in adversarial environments are vulnerable to **deceptive inter-agent communication** from compromised (traitor) nodes.
- Existing multi-agent RL frameworks assume honest communication — they lack mechanisms to detect or suppress false broadcasts.
- Safe swarm coordination breaks down under even a small number of deceptive agents, causing collision cascades and mission failure.
- A principled framework for **trust-aware, resilient swarm navigation** is needed.

OBJECTIVES AND RESEARCH MOTIVATION

Objectives

- Design a trust-aware defense framework for multi-drone coordination.
- Build a continuous-control MARL environment for robust navigation.
- Integrate trusted LiDAR sensing with untrusted peer communication.
- Enable traitor-node detection and adversarial resilience.

Research Motivation

- Autonomous drone swarms are increasingly deployed in contested environments.
- Bio-inspired immune systems offer a proven model for anomaly detection.
- No existing MARL framework combines trust gating with swarm RL in continuous space.

EXPECTED CONTRIBUTION

- **TA-MAPPO framework:** A trust-aware extension of MAPPO with bio-inspired T-cell communication filtering.
- **Trust-gated architecture:** Attenuates deceptive features at the input level rather than penalizing agents post-hoc.
- **Social-distancing reward:** A novel reward shaping mechanism for congestion resolution in continuous swarm RL.
- **Validated environment:** A PettingZoo-based 2D swarm simulator with obstacle navigation and adversarial agent support.

METHODOLOGY AND SYSTEM DESIGN

PROPOSED APPROACH

- **Core method:** Trust-Aware Multi-Agent Proximal Policy Optimization (TA-MAPPO).
- **Domain:** Decentralized drone swarm navigation in a bounded 20×20 continuous 2D environment.
- **Core challenge:** Maintaining safe, coordinated behavior under deceptive or unreliable inter-agent communication.
- **Learning paradigm:** Centralized Training with Decentralized Execution (CTDE).

- **Sensor Input:** Each agent observes trusted LiDAR signals and receives neighbor communication vectors.
- **Trust Evaluation:** A bio-inspired T-cell mechanism estimates per-neighbor trust scores T_{ij} .
- **Feature Gating:** Incoming communication filtered via the trust mask $h_j \odot T_{ij}$ before policy input.
- **Policy Learning:** A PPO-based actor-critic maps trust-filtered observations to control actions.
- **Action Output:** The policy emits continuous thrust commands (v_x, v_y) for each drone.

ALGORITHM AND WORKFLOW

1. **Observe:** Each drone samples local LiDAR rays and receives N neighbor message vectors.
2. **Assess Trust:** The T-cell module evaluates communication consistency and computes T_{ij} .
3. **Gate Features:** Low-trust neighbor representations are attenuated prior to policy inference.
4. **Select Action:** The decentralized PPO actor outputs continuous velocity commands.
5. **Update Policy:** Rewards, value estimates, and advantage terms jointly optimize actor and critic.

Simulation Framework

- **PettingZoo ParallelEnv** — multi-agent environment API.
- Custom VecEnv wrapper for Stable-Baselines3 compatibility.
- Bounded 20×20 continuous 2D space with smooth drone dynamics.
- Time step $\Delta t = 0.1$ s; 800-step episode cap.

Software Stack

- **Python** — primary implementation language.
- **Stable-Baselines3** — PPO training backbone.
- **PyTorch** — custom MLP actor and critic networks.
- **NumPy / Matplotlib** — environment logic and result visualization.

IMPLEMENTATION PROGRESS

MODULES DEVELOPED

- **Swarm Environment:** Continuous 20×20 PettingZoo-based multi-drone simulator.
- **Obstacle Engine:** Graph-based pathfinding with solvability validation for obstacle-map generation.
- **Trust Evaluator:** Bio-inspired T-cell module for per-neighbor communication reliability scoring.
- **Validation Suite:** Automated 5-fold statistical framework for reproducible performance analysis.

INTEGRATION STATUS AND COMPLETION

- **Phase A (100%):** Baseline swarm convergence without obstacles or adversaries — completed and validated across multiple seeds.
- **Phase B (80%):** Static obstacle navigation, congestion handling, and synchronization — in progress.
- **Phase C (0%):** Deceptive traitor behavior and trust-score mechanism activation — pending.
- **Phase D (0%):** Aggressive adversarial modeling and full TA-MAPPO defensive evaluation — pending.

SIMULATION SETUP

ENVIRONMENT AND PARAMETERS

Environment

- Bounded 20×20 continuous 2D navigation space.
- LiDAR sensing: trusted, un-spoofable geometric sensing.
- Inter-drone communication: untrusted, adversarially modeled.
- Velocity and acceleration bounded by safety constraints.

Key Parameters

- **RL algorithm:** PPO with linear LR decay.
- **Learning rate:** 5×10^{-5} .
- **Entropy coeff.:** 0.005 (early); 0.03 (hardening).
- **Rollout workers:** 10 CPU processes, 10 drones each.
- Reward normalization **on**; observation normalization **off**.

TEST SCENARIOS AND PERFORMANCE METRICS

Test Scenarios

- **Random Spawn:** Sparse initial positions across the arena.
- **Dense Cluster:** Congested initial configuration stress-testing collision avoidance.
- **Obstacle Maps:** Diverse, solvability-validated layouts (Phase B).
- **Adversarial:** Traitor agents broadcasting false data (Phases C & D).

Performance Metrics

- **Success Rate:** Fraction of drones reaching the target within the allowed steps.
- **Collision Rate:** Frequency of inter-drone proximity violations.
- **Time to Goal:** Mean steps required for swarm convergence.

RESULTS

CTDE FRAMEWORK — CRITIC (GLOBAL STATE)

Training paradigm: Centralized Training with Decentralized Execution (CTDE).

The **critic** sees the full global state during training to compute accurate advantage estimates:

- Each drone contributes a **52-feature block**: global (x, y) , (v_x, v_y) , and full 48-feature LiDAR sweep.
- For a 10-drone swarm: $10 \times 52 = 520$ total critic input features.
- Global view is *only* used during training — never at execution time.

State Block	Features	Contents
Per Drone $\times 10$	52	Global pos, vel + 48-feature LiDAR
Total	520	Global state for Advantage computation

CTDE FRAMEWORK — ACTOR (LOCAL OBSERVATIONS)

The **actor** operates solely on each drone's local observations at both training and execution time:

Feature Category	Amt	Description
Self-Kinematics	6	Velocity (v_x, v_y) , unit vector to goal (u_x, u_y) , normalised distance, heading ($\arctan 2$).
LiDAR Sweep	48	16 radial sectors $\times$ 3 values: Min (nearest obstacle), Mean (clearance), StdDev (complexity).
Neighbor Clusters	63	9 closest neighbors $\times$ 7 features: relative pos, vel, visibility flag, comm-link status.
Temporal Context	13	Last 5 positions (loop detection) + congestion metrics + trust anomaly scores.
Total	130	Local features for real-time flight control.

PHASE A RESULTS & COMPARATIVE ANALYSIS

Phase A Results

Scenario	Success	Std Dev
Random Spawn	99.68%	± 0.19
Dense Cluster	95.78%	± 0.42

Highly stable convergence across both sparse and congested scenarios.

Comparative Analysis

- Vanilla MAPPO degrades in dense settings due to swarm panic and collision cascades.
- TA-MAPPO's social-distancing reward resolves congestion without destabilizing training.
- Dense-cluster rate of **95.78%** surpasses the 89.45% ceiling of vanilla MAPPO.

NOVELTY AND RESEARCH CONTRIBUTION

RESEARCH CONTRIBUTIONS

- **Bio-inspired trust mechanism:** T-cell anomaly detection adapted for filtering unreliable multi-agent communication — a novel application to swarm RL.
- **Trust-gated neural architecture:** Proactively attenuates deceptive features at the input level rather than reactively penalizing malicious agents.
- **Social-distancing reward:** Resolves physical congestion in continuous-space swarm RL without destabilizing the value landscape.
- **Validated CTDE simulator:** A reproducible, open-architecture PettingZoo environment supporting adversarial phases and 5-fold statistical evaluation.

PUBLICATION READINESS

PUBLICATION PLAN

Target and Focus

- **Venues:** IEEE conferences and workshop tracks on multi-agent learning or autonomous systems.
- **Tentative title:** *TA-MAPPO: Trust-Aware Multi-Agent Reinforcement Learning for Resilient Drone Swarms.*

Proposed Paper Outline

1. Introduction and motivation
2. Related work
3. TA-MAPPO framework and trust model
4. Experimental setup and results
5. Ablation studies on trust gating
6. Conclusion and future work

Additional experiments needed: Adversarial Phase C/D results, trust-score ablation studies, and comparison against state-of-the-art robust MARL baselines.

CHALLENGES AND FUTURE WORK

CHALLENGES & FUTURE WORK

Current Challenges

- Resolving swarm congestion in dense configurations without destabilizing the reward function.
- Generating diverse yet provably solvable obstacle maps.
- Balancing physical realism, training efficiency, and stable RL convergence simultaneously.

Planned Improvements & Timeline

- **Phase C (near-term):** Deceptive traitors broadcasting false positional data.
- **Phase D (mid-term):** Full TA-MAPPO adversarial defensive evaluation.
- **Long-term:** Transition to realistic 3D swarm environments.

THANK YOU